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E-411 DISCONNECTING AND GROUNDING SWITCHES

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installations, commissioning and testing of 500 kV Disconnecting and Grounding Switches in accordance with Part 4 Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirement of Part 4, section 2.5 Codes and Standards and the following specific standards:

2.1 Indonesian National Standards

PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60060 : High-voltage test techniques

IEC 60137 : Insulated bushing for alternating voltages above 1000 V

IEC 60265 : High-voltage switches

IEC 60617 : Graphical symbols for diagrams

IEC 62271 : High-voltage switchgear and controlgear

2.3 National Electrical Manufacturers Association

NEMA 107 : Method of Measurement of Radio Influence Voltage (RIV) of High Voltage Apparatus

2.4 American Society for Testing and Materials

ASTM A-123 : Standard Specification for Zinc (Hot-Dip Galvanized) Coating Steel Products

ASTM A-153 : Specification for Zinc (Hot-Dip) on Iron and Steel Hardware

3. OPERATING AND DESIGN CONDITIONS

The disconnecting and grounding switches shall be designed for outdoor operation 500 kV open air substation located in a tropical climate along the seacoast.

4. DETAILED REQUIREMENTS

4.1 General

The main high voltage current carrying parts of the disconnecting switches shall be able to carry the rated current continuously without exceeding the temperatures specified in IEC 62271.

Full contact and current carrying capacity shall be maintained during reasonable over-travel and under-travel of the mechanism.

All switches of the same rating and service shall be physically and functionally interchangeable.

The completely assembled switch shall be so designed that when the open switch is subject to impulse, 50 Hz, or switching surge voltage (when a switching surge rating is called for in IEC 62271), the dielectric breakdown shall occur to ground and not across the open switch gap nor to adjacent phases.

Grounding switches, when required, shall be of the same short-time current and voltage withstand ratings (across the isolating distance) as the disconnecting switch phase-to-ground ratings.

Switches shall have each blade individually counter-balanced to decrease operating effort and to prevent the blade from falling closed from any open position, even if a part fails or is removed.

Switches shall be designed so that all three (3) blades are in positive continuous control throughout the entire cycle of operation. The three (3) poles shall be in approximate angular synchronizing during the closing operation.

4.2 Ratings

The 500 kV disconnecting switches, complete with insulators and accessories, shall have the following ratings. Each rating shall be assigned on the basis of complete and adequate tests.

- (1) Nominal voltage500 kV
- (2) Rated maximum voltage550 kV
- (3) Frequency50 Hz

- (4) Number of poles 3
- (5) Current ratings:
- a. Continuous current 3,150 A
 - b. Momentary current 100 kA rms
 - c. Short time (1 second) current 40 kA
- (6) Phase-to-ground insulation:
- a. Impulse withstand voltage (1.2 x 50 μ s) crest 1,550 kV
 - b. 50 Hz, dry (and wet*) withstand test voltage 620 kV
 - c. Switching impulse withstand test voltage (class B) 1,175 kV
- (7) Open gap insulation:
- a. Impulse withstand voltage (1.2 x 50 μ s) crest 1,550(+ 315) kV
 - b. One minute, 50 Hz, withstand test voltage 800 kV
 - c. Switching impulse withstand test voltage (class B) 900(+450) kV
- (8) Radio Influence Voltage (RIV)
- The maximum RIV at 1 MHz shall be:
- a. Applied voltage phase-to-ground 335 kV
 - b. R.I.V. 2,500 μ volts
- (9) Control Voltage 220 VDC
- (10) Motor Operating Voltage 220 VDC
- Measurement shall be made in accordance with NEMA 107 procedure. In making the RIV test, radio noise readings shall be taken at various points so as to determine the break point.

4.3 Operating Mechanism

The 500 kV disconnecting switches shall be furnished with single pole operating mechanism. Mechanisms shall be furnished complete with shafting, coupling, crank hanger, etc. and shall have provisions for padlocking in the open and closed positions, with a position indicator located at the operating handle. The operating mechanism can be vertical type or knee type.

When used, each rotating insulator stack shall be provided with a weatherproof

type, greaseless, sealed, ball or roller bearing. All pins, set screws, and washers shall be non-corroding.

Operating rods shall be equipped with weatherproof ball or roller bearings. All bearings shall be provided with brass bushing or other means of avoiding corrosion.

Each single-pole switch unit shall have two (2) adjustable stops at the base, one (1) to block the fully-closed position, and the other to block the fully-open position of the switch blade.

Vertical operating shaft shall be provided with a flexible copper braid connection for grounding the operating mechanism. If vertical operating shaft rotates more than 180 degree, a multi-revolution device shall be furnished for connection to ground grid. Connection shall have an ampacity not less than that of 240 mm² copper cables. Standards pad shall be furnished on the mechanism for ground connection.

4.4 Manual Operation

Manual operation device shall be furnished for all 500 kV disconnecting switches and grounding switches operable from ground level, located approximately one (1) meter above ground with external position indication.

The force to open or close a single-pole or three-pole switch manually shall not exceed 15 kg on a crank not over 0.5 meter in length.

Each grounding switch shall be mechanically interlocked to prevent its closing unless the respective disconnecting switch is fully open. Similarly, each disconnecting switch having an associated grounding switch shall be mechanically interlocked to prevent its closing unless the associated grounding switch is fully open.

4.5 Motor Operation

All 500 kV disconnecting switches shall be motor operated with auxiliary switches and manual operating means. Manual operating means shall meet the requirements of clauses 4.3 and 4.4 of this Specification. Motors shall be 220 V DC with an operating range of **154-253 volts**.

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The motor-driven operating mechanism shall be enclosed with removable doors to

provide access for maintenance and inspection. The enclosure shall have thermostatically controlled space heaters for moisture condensation removal.

Single-pole motor/control circuits, shall be electrically interlocked so that the three (3) single-pole switches will open and close simultaneously upon the closure of a single external control contact (one (1) contact for opening and one (1) contact for closing).

The motor shall be electrically uncoupled whenever the manual switch is engaged. The motor shall be capable of being uncoupled from the switch-operating shaft.

The Contractor shall supply one (1) 220 volt AC, single-phase power supply for the thermostatically controlled heaters.

Each motor operator shall be provided with the following:

- (1) Seal-in relays to permit operation from momentarily closed contacts for disconnecting switch to operate to fully closed or open position.
- (2) Mechanical operation counters within the mechanism housing . Mechanism housing shall be provided with a window for easy viewing of the operation counter.
- (3) Fused disconnecting switches for the heaters and operating mechanism control circuit, each distinctively colored so that the function can be identified.

All disconnecting switches shall be interlocked so that they cannot be operated unless the respective circuit breakers are open. Each disconnecting switch with an associated grounding switch shall be electrically interlocked to prevent its closing unless the associated grounding switch is fully open.

4.6 Auxiliary Switches

Auxiliary switches shall be located in weatherproof cabinets with thermostatically controlled space heaters and they shall be as to be accessible from ground level.

The auxiliary switches shall be field convertible from “a” to “b” vice versa. They shall be capable of interrupting the locked-rotor current of the motor operator, or 30 amperes, whichever is the larger, including four (4) spare auxiliary switches.

Auxiliary switches shall indicate the position of the switchblades and shall be independently of the motor operation.

Sufficient spare shall be provided. All auxiliary switches shall be completely wired to terminal blocks, whether used or not.

4.7 Wiring

Control wiring shall be in accordance with Technical Specification E-280. Power wiring shall be in accordance with Technical Specification E-001.

4.8 Contact Surfaces and Current Carrying Materials

Contacts shall be of the high-pressure contact type and shall open and close with sufficient positive wiping action to remove any film or oxide, but shall not cause galling of the contact surfaces.

The contact fingers at the jaw and shall be adjustable for alignment, and engagement shall be equally divided and each side of the blade, and shall be uniform in size, type and pressure. Contact fingers or permanent shunts at the hinge and shall meet the same requirements.

All movable or sliding contacts (as distinguished from solidly bolted contacts) shall have silver-to-copper or silver-to-silver contact surfaces. Silver shall be inlay or insert type; plating will not be acceptable, except for completely sealed contacts.

Transfer of current beyond the blade contacts through threads or through concealed or semi concealed permanent wiping contacts will be permitted only if the assembly is completely sealed.

All metal parts in contact with line potential (except insulator caps, which may be malleable iron), shall be of non-ferrous metal or stainless steel. All insulator caps and grounded parts of the switch, including the operating mechanism, shall be hot-dipped galvanized after fabrication, in accordance with the latest ASTM Standards A-123 and A-153 or equivalent.

For assembling the live parts, bolts and pins shall be silicon-bronze, copper silicon alloy or stainless steel if in contact with copper or bronze. They shall be alumi-

um alloy or stainless steel, if in contact with aluminium. For bolting aluminium to tinned copper or bronze, stainless steel bolts shall be used.

4.9 Insulators

Supporting and operating rod insulators shall conform to the requirements in Technical Specifications E-001.

4.10 Nameplate

Nameplates shall be furnished in accordance with IEC 62271.

4.11 Corona

Switch and connector surfaces in contact with line potential shall contain no projecting points or irregularities and they shall be designed for a minimum of visible corona or brush discharge when the switch is completely assembled.

No visual corona shall exist in either the open or closed switch position at the applied voltage. Also, the lights shall be turned off in the test bay and visual corona start and stop voltages to be recorded with the corresponding radio noise voltages on both ascending and descending voltage values. Photographs shall be taken for each recorded voltage value. NEMA 107 test procedures shall be used.

4.12 Electrical Clearances

The minimum phase and ground distance clearances are in Technical Specification E-401.

5. PERFORMANCE TESTING AND GUARANTEES

Normal type tests as per IEC 62271 shall have been performed on a sample of each type of switch or switches of identical design and the test results shall be submitted.

Special type tests as per IEC 62271 shall be performed on a sample of each type of switch or have been performed on switches of identical design for radio influence level and corona tests specified in Clause 4.11 of this Specification.

If the type test on identical sample has been performed than test certificate shall be submitted, and no repeated test is required.

Routine tests as per IEC 62271 shall be carried out as outlined below.

- (1) Power frequency voltage dry tests of the main circuit of each voltage level shall be performed on 10 % of each design units with minimum of two (2) units tested.
- (2) Voltage tests on control and auxiliary circuit shall be performed on all units.
- (3) Measurements of the resistance of the main circuit samples are same as clause above (1) “power frequency voltage dry tests” of this Specification.
- (4) Mechanical operating tests shall be performed on all units. If complete assembling of the disconnecting switch is not possible at the manufacturer's works, the operator mechanism shall be loaded to simulate disconnecting mechanism normal loading.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control shall conform to the requirements included in Part 4, Section 2 Subsection 2.7.

The quality assurance/quality control procedures specified above shall include, but not be limited to, the following:

- (1) Power frequency voltage withstand tests.
- (2) Voltage tests on wiring.
- (3) Resistance measurement tests on main circuit.
- (4) Mechanical operating tests.
- (5) Special type tests performed for this Specification.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the following:

- (1) Shop Test Reports.
- (2) All other documentation required by applicable codes and standard.

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